Boolean Algebra

Define Boolean Expression

Boolean expressions are the expressions that evaluate a condition and result in a boolean value i.e. true or false.

example; $a > b$ && $a > c$ is a boolean expression. ~~test~~.

Boolean function

~~A function from B^n~~

Let $B = \{0, 1\}$. Then $B^n = \{(x_1, x_2, \ldots x_n) \mid x_i \in B \text{ for } 1 \leq i \leq n\}$ is the set of all possible n-tuples of 0 and 1.

The variable x is called boolean variable if it assumes values only from B.

A function from B^n to B is called a Boolean function of degree n. if a Boolean expression of n variables can specify it.

Find the value of the Boolean expression:

$$F(x, y, z) = xy + \bar{z}$$

The values of this fuction are displayed in the table.

x	y	z	$\bar{z}$	xy	$xy+\bar{z}$
1	1	1	0	1	1
1	1	0	1	1	1
1	0	1	0	0	0
1	0	0	1	0	1
0	1	1	0	0	0
0	1	0	1	0	1
0	0	1	0	0	0
0	0	0	1	0	1

* law গুলো : 851 page 9

prove the absorption law, $x(x+y) = x$

L.H.S.

$x(x+y)$

$= (x+0)(x+y)$ [Identity law for the Boolean sum]

~~$= (x+0)y$~~

$= x + 0 \cdot y$ [Distributive law of Boolean sum over the Boolean product]

$= x + y \cdot 0$ [commutative law for Boolean product]

$= x + 0$ [Domination law for Boolean product]

$= x$ [Identity for Boolean sum]

$=$ R.H.S. [Proved]

Find the boolean product of $A=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$, $B=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 0\end{bmatrix}$

$$A\odot B=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\odot\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 0\end{bmatrix}$$

$$=\begin{bmatrix}(1\wedge1)\vee(0\wedge0) & (1\wedge1)\vee(0\wedge1) & (1\wedge0)\vee(0\wedge0)\\ (0\wedge1)\vee(1\wedge0) & (0\wedge1)\vee(1\wedge1) & (0\wedge0)\vee(1\wedge0)\end{bmatrix}$$

$$=\begin{bmatrix}1\vee0 & 1\vee0 & 0\vee0\\ 0\vee0 & 0\vee1 & 0\vee0\end{bmatrix}$$

$$=\begin{bmatrix}1 & 1 & 0\\ 0 & 1 & 0\end{bmatrix}$$

Define Duality.

Duality states that the dual of a Boolean expression is obtained by interchanging Boolean sums and Boolean products and interchanging 0s and 1s.

Find the dual of $x(y+0)$ and $\bar{x}.1+(\bar{y}+z)$

The dual can be obtaind by interchanging sum and product and interchanging 1 and 0.

$\therefore$ $x(y+0)$ becomes $x+(y\cdot1)$

$\bar{x}\cdot1+(\bar{y}+z)$ becomes $(\bar{x}+0)(\bar{y}\cdot z)$

Define Boolean Algebra:

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What do you mean by SOP and POS ?

SOP:

SOP stands for sum of product. Of SOP form is a set of product terms that are summed together. It is also known as minterm.

POS:

POS stands for product of sum. POS form is a set of sum terms that are producted together. It is also known as maxterm.

Find the SOP expansion:

$$F(x,y,z) = (x+y)\bar{z}$$

We can construct the SOP expansion by determining the values of F for all possible values of the variable x, y and z.

x	y	z	$\bar{z}$	$x+y$	$(x+y)\bar{z}$
1	1	1	0	1	0
1	1	0	1	1	1
1	0	1	0	1	0
1	0	0	1	1	1
0	1	1	0	1	0
0	1	0	1	1	1
0	0	1	0	0	0
0	0	0	1	0	0

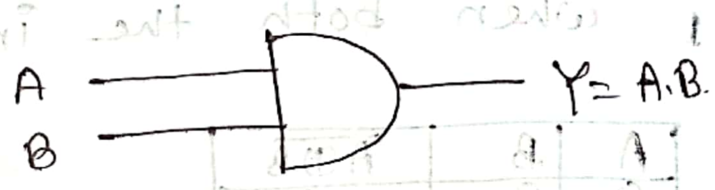
The SOP expansion of F is the Boolean sum of ~~product~~ three minterms corresponding to the three rows of this table that give the value 1 the function. That gives,

$$F(x,y,z) = xy\bar{z} + x\bar{y}\bar{z} + \bar{x}y\bar{z}$$

Truth table and circuit for AND, OR, NOT, Ex-OR, Ex-nor gates.

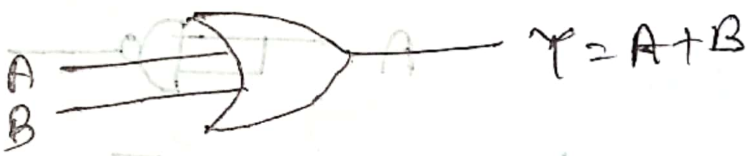
AND Gate: In AND Gate, the output will be 1 if and only if the inputs are 1.

A	B	Y = A.B
1	1	1
1	0	0
0	1	0
0	0	0



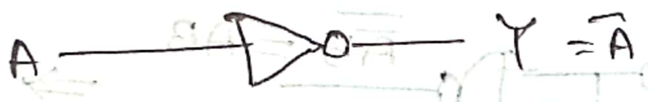
OR Gate: The output of OR gate will be 1 if one or more inputs attain the state 1.

A	B	Y = A+B
0	0	0
0	1	1
1	0	1
1	1	1



NOT Gate: The output of Not ~~is~~ gate is the logical inversion of input.

A	Y
1	0
0	1

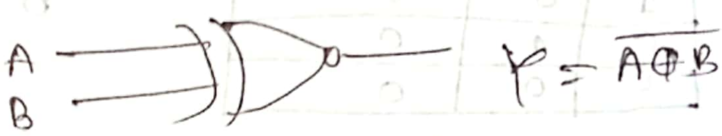


X-OR: The output of X-OR Gate attains 1 if exactly one of the inputs attain 1.

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

X-nor: The output of X-nor gate attains 1 when both the inputs are same state.

A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1



Universality of nand gate:

Nand to NOT:

$Y = \overline{A \cdot A} = \bar{A} \quad [A \cdot A = A]$.

$\therefore$

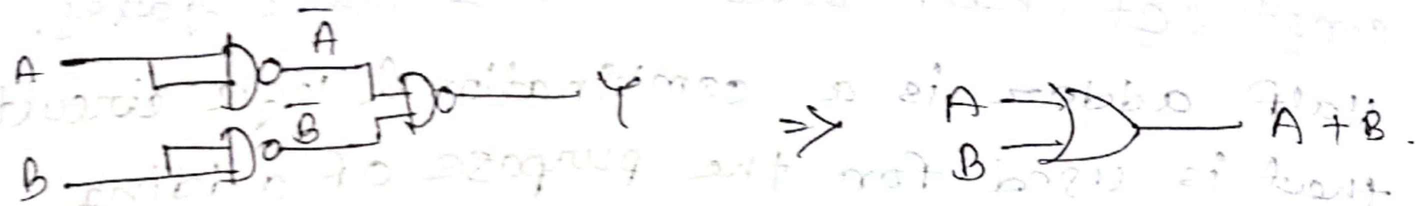
Nand to AND:

$\Rightarrow$

here $Y = \overline{\overline{A \cdot B}}$
$= AB \quad [\because \bar{\bar{A}} = A]$

which is and gate.

Nand to OR:



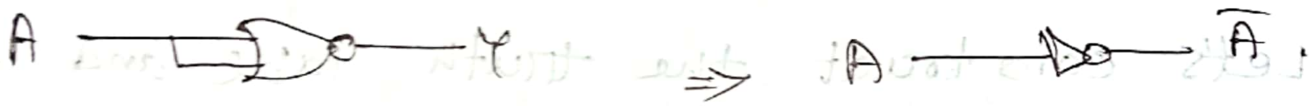
here, $Y = \overline{\bar{A}\cdot\bar{B}} = \bar{\bar{A}} + \bar{\bar{B}}$ $[\overline{A\cdot B} = \bar{A} + \bar{B}]$

$= A + B$ $[\bar{\bar{A}} = A]$

which is or gate.

Universality of OR gate:

NOR to NOT:



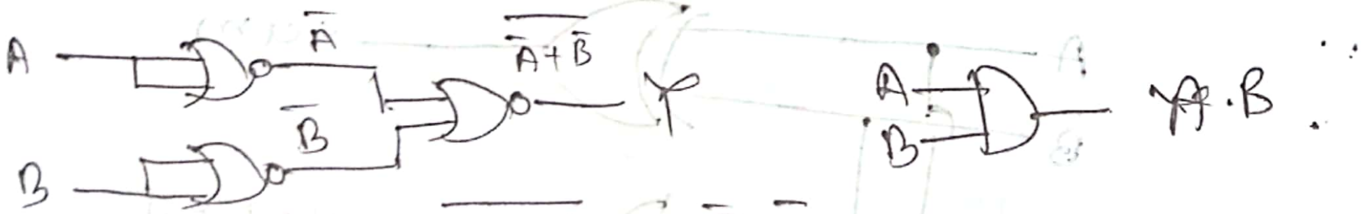
here. $\overline{A+A} = \bar{A}$ $[\because A + A = A]$

NOR to OR:

here. $Y = \overline{\overline{A+B}} = A + B$ $[\bar{\bar{A}} = A]$

which is OR gate.

NOR to and:



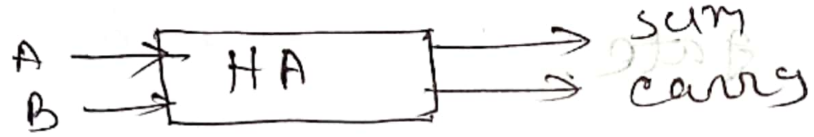
here. $Y = \overline{\bar{A}+\bar{B}} = \bar{\bar{A}}\cdot\bar{\bar{B}}$

$= AB$.

which is AND Gate.

construct half adder using logic gates.

Half adder is a combinational logic circuit that is used for the purpose of adding two single bit numbers. It contains 2 inputs and 2 outputs.



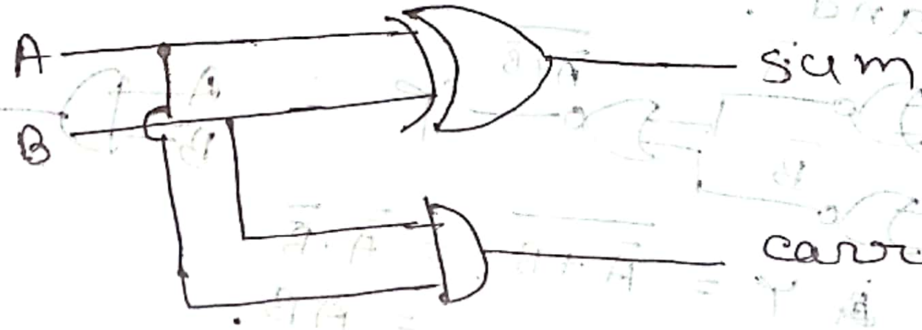
Block diagram.

Let's construct the truth table and simplify the expression:

A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$\therefore$ $sum = \bar{A}B + A\bar{B}$ $\quad$ $carry = AB$.

$= A \oplus B$.



HA logic diagram

Construct full adder using logic gates.

Full adder is a combinational logic circuit that is used for the purpose of adding two single bits with a carry. It contains 3 ~~bits can~~ inputs and 2 outputs.

Let's construct the truth table and simplify the expression :

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$\therefore$ Sum $= \bar{A}\bar{B}C_{in} + \bar{A}BC_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$

$= \bar{A}(\bar{B}C_{in} + B\bar{C}_{in}) + A(\bar{B}\bar{C}_{in} + BC_{in})$

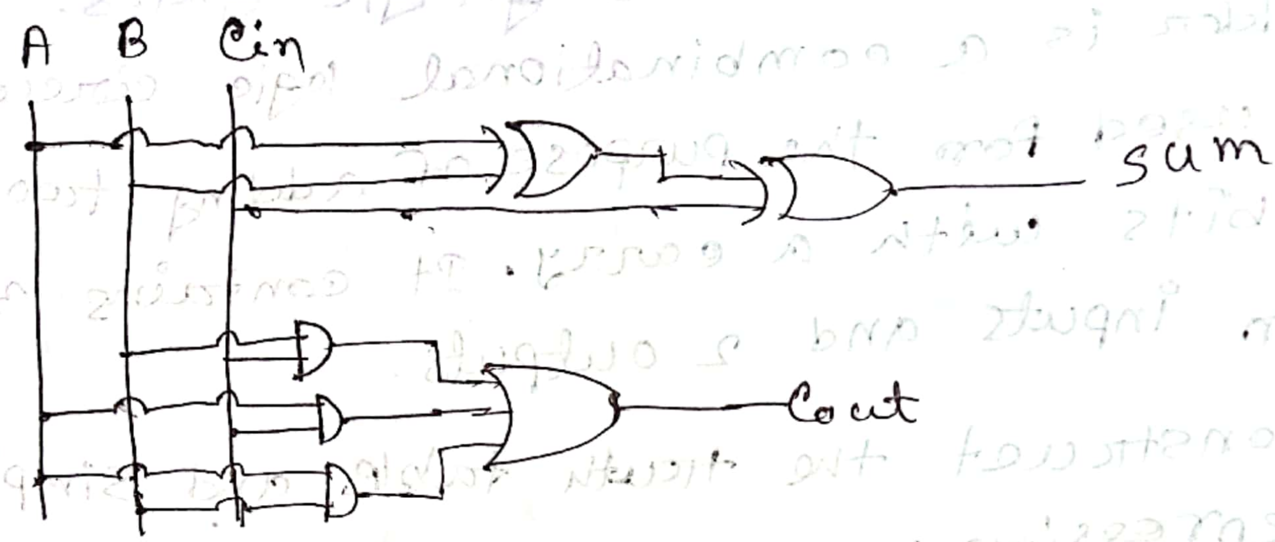
$= \bar{A}(B \oplus C_{in}) + A(\overline{B \oplus C_{in}})$

$= A \oplus B \oplus C_{in}$

Cout $= \bar{A}BC_{in} + A\bar{B}C_{in} + AB\bar{C}_{in} + ABC_{in}$

$= \bar{A}BC_{in} + ABC_{in} + A\bar{B}C_{in} + ABC_{in} + AB\bar{C}_{in} + A$

$= BC_{in} + AC_{in} + AB$



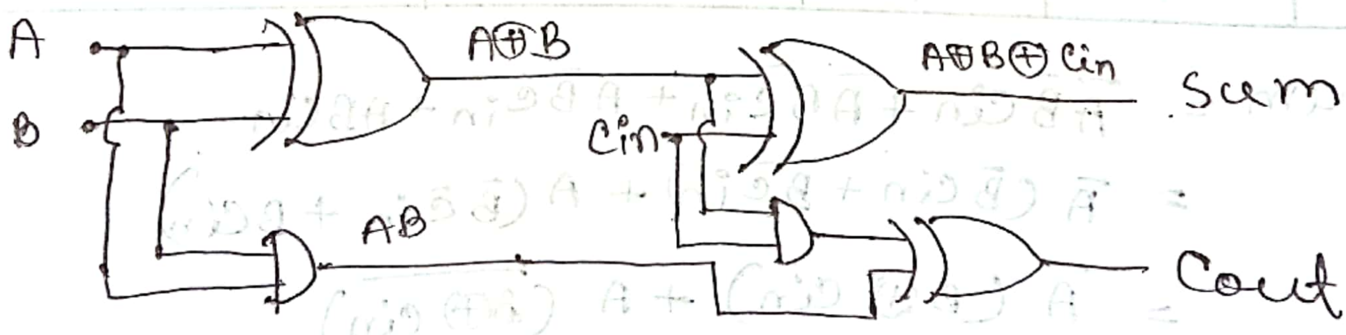
Construct full Adder using half adder.

or from, above,

$$Sum = A \oplus B \oplus Cin$$

$$Cout = \bar{A}B\,Cin + A\bar{B}\,Cin + AB\,\overline{Cin} + AB\,Cin$$

$$= Cin(A \oplus B) + AB$$



Using K-map, minimize the SOP expressions:

~~(i) $ABC + \bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}C$~~

(i) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$

Since the expression has 3 variables, so we need to draw $2^3 = 8$ cells.

x \ yz	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
x		1	1	
$\bar{x}$	1		1	

Here, we have three pairs.

$\therefore F = \bar{x}yz + x\bar{z} + \bar{y}\bar{z}$

(ii) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z$

x \ yz	yz	$y\bar{z}$	$\bar{y}\bar{z}$	$\bar{y}z$
x			1	1
$\bar{x}$	1			1

$\therefore F = x\bar{y} + \bar{x}z$

(i) $\bar{x}\bar{y}z + \bar{x}yz$

$x \backslash yz$	yz	$\bar{y}z$	$\bar{y}\bar{z}$	$y\bar{z}$
x				
$\bar{x}$	1	1		

$\therefore F = \bar{x}z$

(ii) $xy\bar{z} + x\bar{y}z + \bar{x}\bar{y}z + \bar{x}y\bar{z}$

$x \backslash yz$	yz	$\bar{y}z$	$\bar{y}\bar{z}$	$y\bar{z}$
x		1		1
$\bar{x}$	1 1			1

$\therefore F = \bar{x}y + y\bar{z} + x\bar{y}z$

(iii) $xy\bar{z} + x\bar{y}z + \bar{x}yz + \bar{x}\bar{y}z$

$x \backslash yz$	yz	$\bar{y}z$	$\bar{y}\bar{z}$	$y\bar{z}$
x		1		1
$\bar{x}$	1	1		

$\therefore F = \bar{x}z + \bar{y}z + xy\bar{z}$

(iv) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$

$x \backslash yz$	yz	$\bar{y}z$	$\bar{y}\bar{z}$	$y\bar{z}$
x		1	1	
$\bar{x}$	1	1	1	

$F = \bar{y} + \bar{x}z$

(v) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + xyz + \bar{x}\bar{y}\bar{z} + z + x$

$x \backslash yz$	yz	$\bar{y}z$	$\bar{y}\bar{z}$	$y\bar{z}$
x	1	1	1	1
$\bar{x}$	1	1	1	

$F = z + x + \bar{y}$